SERVICE HEADER (Fill Once)

Service ID	Service Name	Dept ID	Version	Status	Total Steps	Est. Duration
SVC-001	[Building Permit]	1	1	DRAFT	4	144 hrs

WORKFLOW STEPS MASTER TABLE (Core Configuration)

Step	Role Code (ID)	Form Code (ID)	Jurisdiction (ID)	Strategy (ID)	Actions (IDs)	Transitions (JSON)	Assignment Rule	SLA Hrs	Escalation	Parallel OK	Notes
1	APPLICANT (3)	AF (1)	-	-	4	{"SUBMIT":{"next step":2}}	{}	**0**	-	❌	Applicant uploads
2	NODAL (7)	PF (2)	DISTRICT (2)	ROLE (1)	1,2,3	{"FORWARD":{"next step":3,"next_roles":[33]},"APPROVE":{"next step":4},"REJECT":{"next step":1}}	{"strategy":"ROLE","role id":7,"targets":[{"role id":33}]}	**72**	Role 15 @48h	✅ F only	Doc verification
3	VERIFIER (33)	VF (3)	DISTRICT (2)	OFFICE (3)	1,2	{"FORWARD":{"next step":4},"APPROVE":{"next step":4}}	{"strategy":"OFFICE","office_ids":[101,102]}	**48**	-	❌	Field inspection
4	APPROVER (8)	PF (2)	STATE (1)	RULE (4)	2	{"APPROVE":{"next step":0}}	{"strategy":"RULE","conditions":[{"field":"amount","op":">","value":100000}]}	**24**	Dept Head	❌	Final decision

MASTER REFERENCE TABLES (Copy Codes Here ↓)

ROLES

Code	ID	Name
APPLICANT	3	Applicant

"slaBreachRequiresReason": true
},
{
"step": 4,
"currentRoleId": "JD_ROLE_ID",
"formTypeId": "PROCESSING_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "",
"nextRoleId": "MD_ROLE_ID",
"forwardRoleId": "MD_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "NODAL_OFFICER_ROLE_ID",
"actionMasterIdsJson": ["F", "R"],
"transitionMapJson": {
"F": { "nextStep": 5, "nextRoleId": "MD_ROLE_ID" },
"R": { "nextStep": 3, "nextRoleId": "NODAL_OFFICER_ROLE_ID" }
},
"slaHours": 24,
"slaBreachRequiresReason": true
},
{
"step": 5,
"currentRoleId": "MD_ROLE_ID",
"formTypeId": "APPROVAL_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "",
"nextRoleId": "MEETING_SECRETARY_ROLE_ID",
"forwardRoleId": "MEETING_SECRETARY_ROLE_ID",
"approverId": "SECRETARY_ROLE_ID",
"revertRoleId": "NODAL_OFFICER_ROLE_ID",
"actionMasterIdsJson": ["A", "R"],
"transitionMapJson": {
"A": { "nextStep": 6, "nextRoleId": "MEETING_SECRETARY_ROLE_ID" },
"R": { "nextStep": 3, "nextRoleId": "NODAL_OFFICER_ROLE_ID" }
},
"slaHours": 24,
"slaBreachRequiresReason": true
},
{
"step": 6,
"currentRoleId": "MEETING_SECRETARY_ROLE_ID",
"formTypeId": "SCHEDULING_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "MEETING_SCHEDULING",
"nextRoleId": "MEETING_SECRETARY_ROLE_ID",
"forwardRoleId": "MEETING_SECRETARY_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "MD_ROLE_ID",
"actionMasterIdsJson": ["F"],
"transitionMapJson": {
"F": { "nextStep": 7, "nextRoleId": "MEETING_SECRETARY_ROLE_ID" }
},
"slaHours": 12,
"slaBreachRequiresReason": true
},{
"step": 7,
"currentRoleId": "MEETING_SECRETARY_ROLE_ID",
"formTypeId": "MEETING_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "RECORD_PROCEEDINGS",
"nextRoleId": "MD_ROLE_ID",
"forwardRoleId": "MD_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "MEETING_SECRETARY_ROLE_ID",
"actionMasterIdsJson": ["F"],
"transitionMapJson": {
"F": { "nextStep": 8, "nextRoleId": "MD_ROLE_ID" }
},
"slaHours": 12,
"slaBreachRequiresReason": true
},{
"step": 8,
"currentRoleId": "MD_ROLE_ID",
"formTypeId": "NOTIFICATION_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "QUERY_NOTIFICATION",
"nextRoleId": "APPLICANT_ROLE_ID",

"forwardRoleId": "APPLICANT_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "MEETING_SECRETARY_ROLE_ID",
"actionMasterIdsJson": ["F", "P"],
"transitionMapJson": {
"F": { "nextStep": 9, "nextRoleId": "APPLICANT_ROLE_ID" },
"P": { "nextStep": 8, "nextRoleId": "MD_ROLE_ID" }
},
"slaHours": 48,
"slaBreachRequiresReason": true
},
{
"step": 9,
"currentRoleId": "APPLICANT_ROLE_ID",
"formTypeId": "RESPONSE_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "APPLICANT_RESPONSE",
"nextRoleId": "NODAL_OFFICER_ROLE_ID",
"forwardRoleId": "NODAL_OFFICER_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "MD_ROLE_ID",
"actionMasterIdsJson": ["F"],
"transitionMapJson": {
"F": { "nextStep": 3, "nextRoleId": "NODAL_OFFICER_ROLE_ID" }
},
"slaHours": 48,
"slaBreachRequiresReason": true
},
{
"step": 10,
"currentRoleId": "APPLICANT_ROLE_ID",
"formTypeId": null,
"permissableTabFormId": "",
"subformActionName": "TERMINATION",
"nextRoleId": null,
"forwardRoleId": null,
"approverId": null,
"revertRoleId": null,
"actionMasterIdsJson": ["C"],
"transitionMapJson": {
"C": { "nextStep": 0 }
},
"slaHours": 0,
"slaBreachRequiresReason": false
}
]

Code	ID	Name
NODAL	7	Nodal Officer
VERIFIER	33	Field Verifier
APPROVER	8	State Approver
DC	15	District Collector

FORM TYPES

Code	ID	Name
AF	1	Applicant Form
PF	2	Processing Form
VF	3	Verifier Form

JURISDICTION

Code	ID	Name
STATE	1	State Level
DISTRICT	2	District Level
BLOCK	3	Block Level

ASSIGNMENT STRATEGY

Code	ID	Name
ROLE	1	Role Based
USER	2	Manual Users
OFFICE	3	Office Based
RULE	4	Conditional Rules

ACTIONS (Use IDs)

Code	ID	Name	Color
FORWARD	1	🔄 Forward	blue
APPROVE	2	✅ Approve	green
REJECT	3	❌ Reject	red
SUBMIT	4	📤 Submit	orange
REVERT	5	🔙 Revert	yellow

SUBFORM ACTIONS

Code	Use
`DOCUMENT_VERIFICATION`	Docs check
`INSPECTION`	Field visit

Code	Use
`TRANSACTION`	Payment/audit
`""`	None

ADVANCED FEATURES SUPPORTED

1. Complex Transitions

```
{"FORWARD":{"next_step":3,"next_roles":[33,15]},"APPROVE":{"next_step":4}}
```

2. Assignment Rules

```
{"strategy":"RULE","conditions":[{"field":"district","op":"=","value":"Dehradun"}]}
{"strategy":"USER","user_ids":[1001,1005]}
```

3. SLA Escalation

Format	Example
`@48h:Role15`	Escalate to Role 15 after 48h
`Head@24h`	To Dept Head after 24h

4. Parallel Processing

Symbol	Meaning
✅ F only	Parallel FORWARD allowed
✅ All	All actions parallel
❌	Sequential only

AUTOMATIC VALIDATION RULES

- ✅ Steps must be sequential (1,2,3...)
- ✅ Every step needs ≥1 action
- ✅ SLA auto-sum for total duration
- ✅ Role/Form codes validated from tables
- ✅ JSON syntax checked

GENERATION COMMANDS

Excel/Sheets → JSON:

```
=FILL_TABLE() → workflow-SVC001-v1.json
```

Expected Output Preview:

CAF Workflow JSON (10 Steps)


```
[
{
"step": 1,
"currentRoleId": "APPLICANT_ROLE_ID",
"formTypeId": "APPLICANT_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "",
"nextRoleId": "NODAL_OFFICER_ROLE_ID",
"forwardRoleId": "NODAL_OFFICER_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "APPLICANT_ROLE_ID",
"actionMasterIdsJson": ["F", "R"],
"transitionMapJson": {
"F": { "nextStep": 2, "nextRoleId":
"NODAL_OFFICER_ROLE_ID" },
"R": { "nextStep": 1, "nextRoleId":
"APPLICANT_ROLE_ID" }
},
"slaHours": 48,
"slaBreachRequiresReason": true
},
{
"step": 2,
"currentRoleId": "MD_ROLE_ID",
"formTypeId": null,
"permissableTabFormId": "",
"subformActionName": "",
"nextRoleId": "NODAL_OFFICER_ROLE_ID",
"forwardRoleId": "NODAL_OFFICER_ROLE_ID",
"approverId": "MD_ROLE_ID",
"revertRoleId": "APPLICANT_ROLE_ID",
"actionMasterIdsJson": ["AUTO"],
"transitionMapJson": {
"AUTO": { "nextStep": 3, "nextRoleId":
"NODAL_OFFICER_ROLE_ID" }
},
"slaHours": 0,
"slaBreachRequiresReason": false
},
{
"step": 3,
"currentRoleId": "NODAL_OFFICER_ROLE_ID",
"formTypeId": "PROCESSING_FORM_ID",
"permissableTabFormId": "",
"subformActionName": "",
"nextRoleId": "JD_ROLE_ID",
"forwardRoleId": "JD_ROLE_ID",
"approverId": "JD_ROLE_ID",
"revertRoleId": "APPLICANT_ROLE_ID",
"actionMasterIdsJson": ["F", "R", "H"],
"transitionMapJson": {
"F": { "nextStep": 4, "nextRoleId": "JD_ROLE_ID" },
"R": { "nextStep": 1, "nextRoleId":
"APPLICANT_ROLE_ID" },
"H": { "nextStep": 3, "nextRoleId":
"NODAL_OFFICER_ROLE_ID" }
},
"slaHours": 24,
```